

OBJECTIVE: Designing a universal shift register using flip-flops and multiplexers.

COMPONENT ANALYSIS:

<u>Name</u>	<u>Quantity</u>
(i) Trainer board	1 piece
(ii) Ic extractor	1 piece
(iii) Ic (DUAL 4x1 MUX) 74153	2 pieces
(iv) Ic (DUAL D-FF) 7474	2 pieces
(v) Magnifier	1 piece
(vi) Handbook	1 piece.
(vii) wires	—

Problem specification: Design and implement a 4-bit universal shift register.

Excitation table:

Input					Output				
CLEAR	clock	S_1	S_L	P_i	A_3	A_2	A_1	A_0	Mode
0	X	X	X	X	0	0	0	0	Asynchronous clear
1	X	0	0	X	A_3	A_2	A_1	A_0	Hold
1	$\uparrow$	0	1	A_{i+1}	I_R	A_3	A_2	A_1	Shift right
1	$\uparrow$	1	0	A_{i+1}	A_2	A_1	A_0	I_L	Shift left
1	$\uparrow$	1	1	A_i	I_3	I_2	I_1	I_0	Parallel load

Circuit Diagrams

